COSC 369 · SOFTWARE ENGINEERING I · MIDTERM PROJECT

FACULTY TEACHING & OFFICE-HOURS SCHEDULER

Project Extension: Prolog Knowledge System & Automation

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New Knowledge-Base Extension

From Schedule Generator to Knowledge System

```
Existing C Scheduler    +    class_schedule.csv , office_hours.csv    →    build_knowledge_base.py

                →    faculty_info.txt , syllabus facts.txt    →    generated_facts.pl
```

Knowledge Sources

- **Combines:**
  - Class schedule data
  - Generated office hours
  - Faculty information
  - COSC 369 syllabus facts

What the Extension Adds

- **Added syllabus information as a new knowledge source**
- **Created build_knowledge_base.py**
- **Converts existing scheduler output into Prolog facts**
- **Produces generated_facts.pl**

Prolog + PySwip Question System

Asking Questions About the Data

```
$ make ask

====================================
 Faculty Schedule Question System
====================================

Type a question.
Type 'help' for examples.
Type 'exit' to quit.

Question:
```

```
Type a question.
Type 'help' for examples.
Type 'exit' to quit.

Question: Who teaches COSC 369?

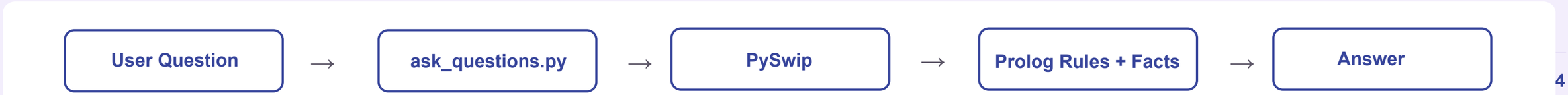
COSC 369 is taught by Dr. Appolo Tankeh.

Question:
```

```
Question: help

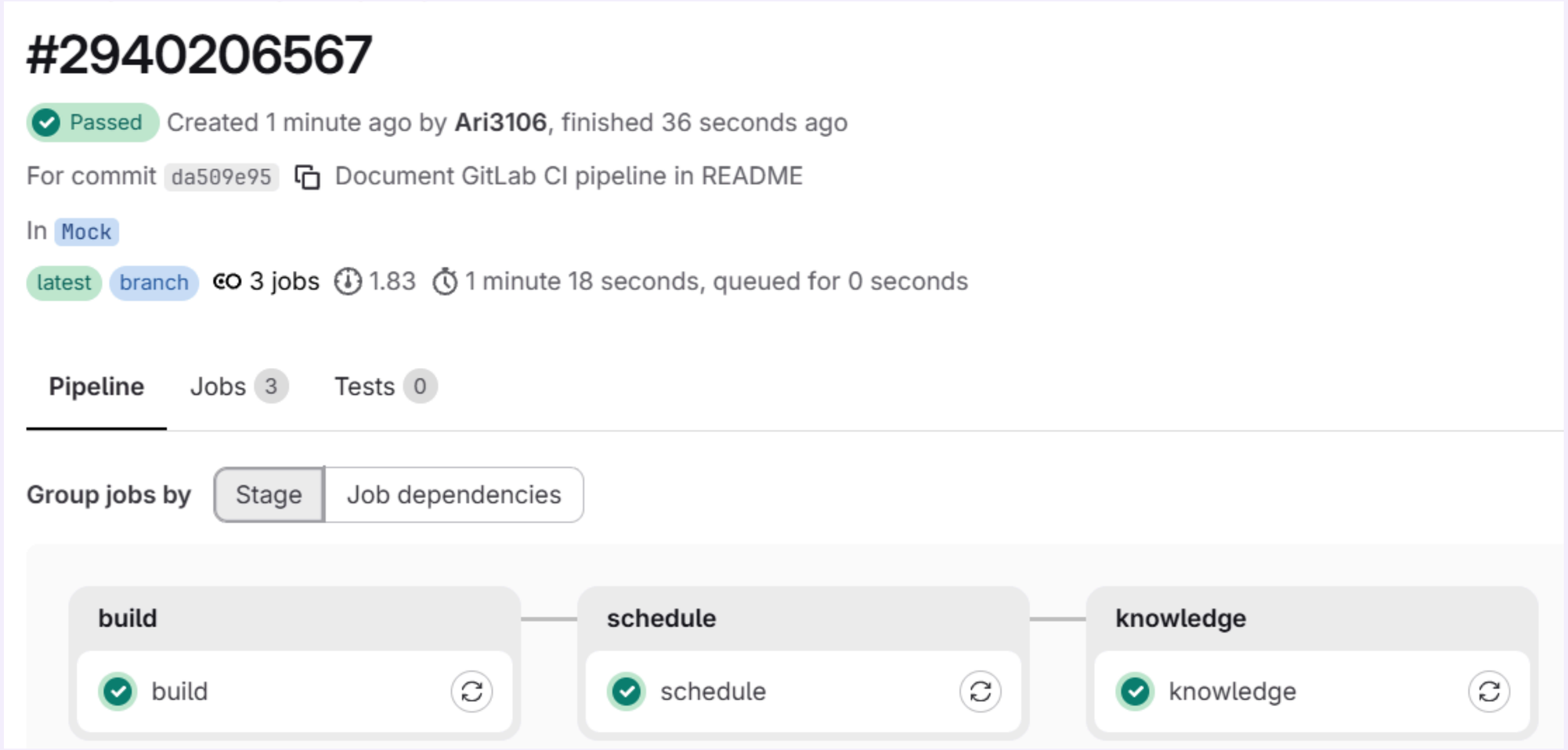
Example questions:

  Who teaches COSC 369?
  What classes does Adedoyin, Anthony teach?
  When are Dr. Appolo Tankeh's office hours?
  Which office hours are on Wednesday?
  Where is Dr. Appolo Tankeh's office?
  What is the email for Dr. Appolo Tankeh?
  What are the prerequisites for COSC 369?
  What is the attendance policy for COSC 369?
  What is the course description for COSC 369?
  When is the midterm for COSC 369?
  When does COSC 369 meet?
```



Automation, Makefile & GitLab CI/CD

Automating the Complete System



Thank You